

NAME: Colleen FretiPERIOD: B**Screen Time, Two Deletions**

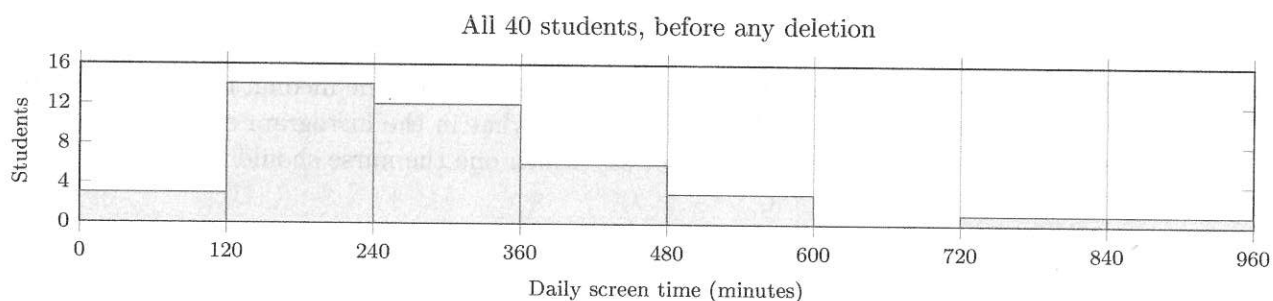
AP Statistics · Unit 1 · Topics 1.1–1.7 (CED 1.1–1.7)

[STAR] THE PROBLEM

A school nurse recorded, for a random sample of 40 students, three things: grade level (Freshman or Senior), the device they use most (Phone, Laptop, or Console), and daily screen time in minutes from the phone's own report. The principal asked one question: *Is too much screen time a concern for our students?* The nurse's first report said only: *12 of the 40 are over 240, that is 30%, so yes.*

Group sizes: 25 freshmen, 15 seniors. Over 240 minutes: 9 freshmen, 3 seniors.

The two largest values in the histogram (a 900-minute and a 720-minute entry) came from two students who logged a whole weekend by mistake. The nurse deleted both and re-computed: the mean fell from 205 to 178 minutes; the median moved from 176 to 174 minutes; the headline became *10 of 38 are over 240, that is 26%.*

**FIRST TAKE — one sentence before you work the parts**

Before any calculation: does the nurse's first sentence answer the principal's question? Name one thing the sentence leaves out.

Nope the nurse's statement doesn't mean much since we don't know what that screen time does to students not only that the statistic is a stark minority

[BOOK] RULES YOU MUST USE (NO ANSWERS HERE)

A number means nothing until you say WHO was measured, WHAT was measured, the UNITS, and the QUESTION it answers.

Individuals are the things measured; variables are the characteristics recorded. A label or ID is not a measurement.

Groups of different sizes are compared with relative frequencies (percents), never raw counts alone.

A full description names shape, center, variability, AND unusual features. Some statistics resist unusual values; some do not.

Work (a)–(d).

DOK 1 (a) Fill the table: who or what are the individuals, and for each recorded variable, what ONE value looks like and whether it is categorical or quantitative. Checklist: an individual is one thing measured; a name, label, or ID is not a measurement; quantitative values could sensibly be averaged.

Individuals:		
Variable	One value looks like	Categorical / Quantitative
Grade level	9th grade	categorical
Device used most	Samsung	categorical
Daily screen time	80000 minutes	quantitative

DOK 2 (b) A senior says: Nine freshmen are over 240 and only three seniors are, so freshmen are three times as bad. Fix the comparison. Frame: Of the 25 freshmen, 9 are over 240, which is 36%. Of the 15 seniors, 3 are over 240, which is 20%. So the fair comparison is misleading because the groups differ. Then explain in one sentence why the count 9 versus 3 can mislead here.

The sentence misleads because the total percent differs the senior judged only the numbers and not total

DOK 2 (c) Deleting the two values moved the mean by 27 minutes and the median by 2 minutes. Which of these two statistics is resistant to unusual values? Explain what in the histogram caused the big move, using the words *resistant* and *unusual values*. Do not say which one the nurse should report yet.

The median is resistant to unusual values that range from 20 to 960 this shifted the mean toward

DOK 3 (d) ★ Judge the nurse's trimmed report in three to five sentences. Before you write, check: (1) WHO, WHAT, and the UNITS behind the 26%; (2) what question 26% answers, and whether that is the principal's question; (3) which center to report, tied to your answer in (c); (4) whether deleting the two mistaken values was responsible and what the report must SAY about them. Use the frame below if it helps.

The 26% are the students who's screen time range from 0 - 600 minutes the percentage acknowledges the all students not including the 2 over 600 this doesn't answer the

Frame: I would report _____ because _____; removing the two values changed _____ but not _____, so the principal should _____.

Turn this sheet in whenever you finish — bonus credit. Part (d) is scored E / P / I.